Chapter test

42 Applications in biotechnology

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following comparisons between traditional breeding and genetic engineering is correct?

	Traditional breeding	*Genetic engineering*
A.	The time taken to generate desirable traits is shorter.	The time taken to generate desirable traits is longer.
B.	The chance of success in producing desirable traits is lower.	The chance of success in producing desirable traits is higher.
C.	The desirable genes are transferred to target organisms more precisely.	The desirable genes are transferred to target organisms less precisely.
D.	Production cost is lower.	Production cost is higher.

2. Which of the following is/are the characteristic(s) of monoclonal antibodies?

(1) They are produced by clones of a single parent B cell.
(2) They are antigen-specific.
(3) They can be easily isolated from the blood of an immunized person.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

★ 3. Microorganisms can be genetically modified to synthesize antigens of pathogens. These antigens can be used

A. to test for pregnancy.
B. to diagnose cancers.
C. to treat infectious diseases.
D. in vaccines to induce immunity against infectious diseases.

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4. Which of the following is ***not*** a characteristic of stem cells?

   A. They are unspecialized cells.
   B. Some types have limited developmental potential.
   C. They undergo limited cycles of meiotic cell division.
   D. They can differentiate into different types of cells. ☐

5. Stem cells could be used for the production of

   A. anti-cancer drugs.
   B. monoclonal antibodies.
   C. organs for transplant.
   D. vaccines. ☐

6. Which of the following statements about somatic cell gene therapy is/are correct?

   (1) All genetic diseases can be treated by somatic cell gene therapy.
   (2) The offspring of the patient receiving somatic cell gene therapy will not have the genetic disease.
   (3) It involves transgenic technique.

   A. (1) only
   B. (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

★ 7. Which of the following diseases is the ***least*** likely to be treated by gene therapy?

   A. influenza
   B. cystic fibrosis
   C. cancer
   D. severe combined immunodeficiency disease ☐

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8. Which of the following is/are the limitation(s) or potential hazard(s) of gene therapy?

(1) The introduced genes may not be integrated into desired positions on the human chromosomes.
(2) Genetic diseases caused by multiple gene defects are too complicated to be treated with current technologies.
(3) Genetic modification of human characteristics arouses ethical concerns.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

9. Which of the following is/are the reason(s) for growing GM crops?

(1) to increase crop yields in order to support the world population
(2) to produce transgenic crops that can be used as vaccines to induce immunity against pathogens
(3) to induce antibiotic resistance in wild plants

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

10. Which of the following statements about Bt crops are correct?

(1) They produce Bt toxins to kill the larvae of pests.
(2) The Bt toxins produced do not kill adult pests.
(3) They may promote the emergence of Bt-resistant insects.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	42.4	42.1	42.1	42.3	42.3	42.2	42.2	42.2	42.4	42.4

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B. Problem solving (22 marks)

11. Genotropin is a synthetic human growth hormone produced using recombinant DNA technology.

(a) Suggest ***one*** medical use of Genotropin. (1 mark)

(b) In the production of Genotropin using recombinant DNA technology, *Escherichia coli* are used as host cells. Describe the main steps of modifying *E. coli* to synthesize the human growth hormone. (5 marks)

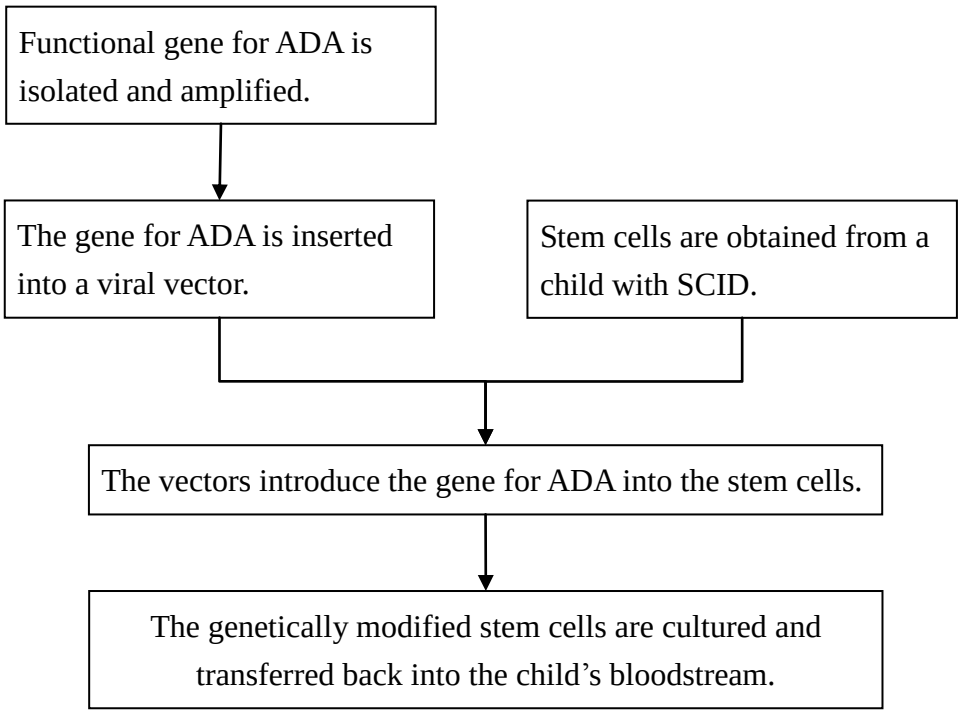
(c) State ***two*** advantages of producing Genotropin by genetic engineering. (2 marks)

(Total: 8 marks)

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★ 12. Severe combined immunodeficiency disease (SCID) is a genetic disease. A type of SCID is caused by a defect in the gene coding for an enzyme called ADA, which is essential for the normal development of white blood cells. Lacking ADA and functional white blood cells, the immune system is weak and cannot defend even common infections.

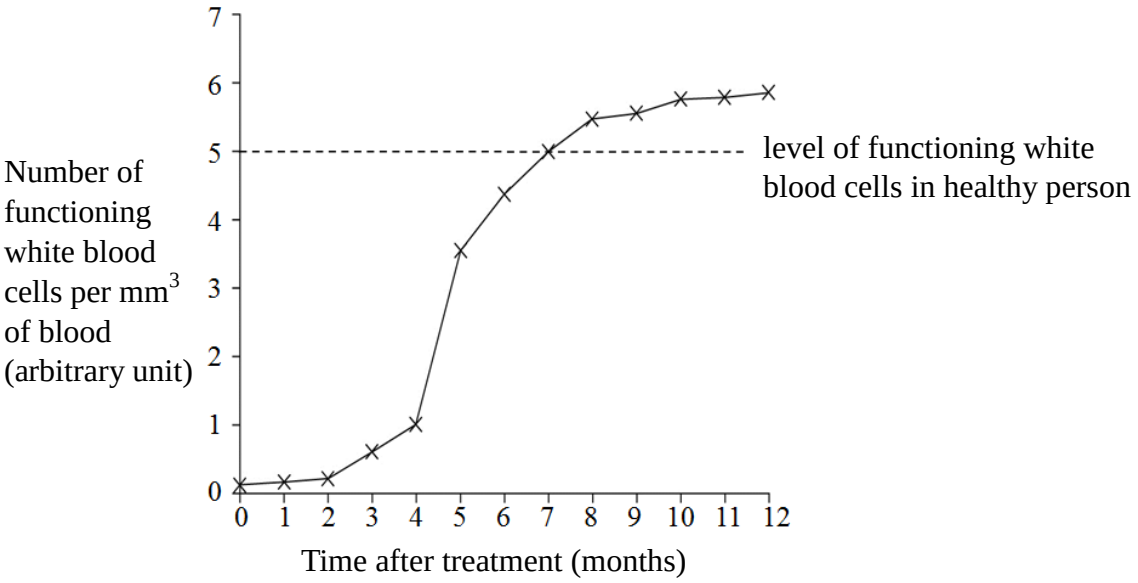
Children with this type of SCID could be treated with stem cells and gene therapy.



(a) From which part of the child's body should stem cells be collected for treatment? Explain your answer. (2 marks)

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(b) A child was treated with the genetically modified stem cells. The number of functioning white blood cells was monitored for a year after the treatment. The graph below shows the results.



Describe and explain the changes in the number of functioning white blood cells during the year. (3 marks)

(Total: 5 marks)

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★★ 13. Pigs normally cannot efficiently digest and absorb the phosphate in their feed (such as maize, legumes and soybeans). The phosphate content in their faeces is high. Scientists have produced genetically modified pigs with a gene coding for the enzyme phytase inserted. The GM pigs synthesize phytase which is secreted in saliva. Phytase catalyses the hydrolysis of phosphate in plant tissues.

(a) Describe an environmental problem that may be caused by the discharge of pig faeces rich in phosphate into rivers. (3 marks)

(b) The scientists produced the GM pigs described above with the method microinjection. Outline the major steps involved. (4 marks)

(c) The GM pigs could help solve the environmental problem caused by pig farming. Suggest ***two*** benefits for farmers who rear these GM pigs. (2 marks)

(Total: 9 marks)

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C. Exam challenge (8 marks)

★★ 14. Herceptin is a monoclonal antibody. It shows high affinity to the receptor HER2, which is commonly found on the surface of many human breast cancer cells. Herceptin is used to treat breast cancer.

In an investigation on breast cancer treatment, the ability of Herceptin and X-ray to induce cell death was measured.

The results are shown in the table below.

Treatment	No treatment	Herceptin only	X-ray treatment only	Herceptin and X-ray treatment
Ability to induce cell death (arbitrary unit)	0.5	0.6	0.7	2.0

(a) What is a monoclonal antibody? (1 mark)

(b) With reference to the results, compare the effects of the 'Herceptin only', 'X-ray only' and 'Herceptin and X-ray' treatments on breast cancer cells. (3 marks)

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(c) Compare the effectiveness of using 'Herceptin and X-ray treatment' with 'Herceptin only treatment' by calculating the percentage difference in the ability to induce cell death. Show your working. (2 marks)

(d) Suggest the role of Herceptin in treating breast cancer. (2 marks)

(Total: 8 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	B	2.	B	3.	D	4.	C	5.	C
6.	B	7.	A	8.	D	9.	B	10.	D

(1 mark each)

B. Problem solving (22 marks)

11. (a) To treat dwarfism / growth hormone deficiency (1)

(b) The gene coding for human growth hormone (HGH) is isolated from human genome. (1)
The gene coding for HGH and plasmids are cut with the same restriction enzyme. (1)
The gene coding for HGH is inserted into the cut plasmid by DNA ligase. (1)
The recombinant plasmid is inserted into *E. coli*. (1)
The bacteria express the gene and synthesize human growth hormone. (1)

(c) Genotropin can be mass-produced in a pure form (free of pathogens). /
Genotropin can be mass-produced at a faster rate. /
Genotropin can be mass-produced at a lower cost. (Any ***two***, 2)

(Total: 8 marks)

12. (a) bone marrow (1)
Stem cells from the bone marrow can differentiate into different types of blood cells. (1)

(b) There were few functional white blood cells in the first four months. Afterwards, the number of functional white blood cells increased drastically. (1)
It takes time for the stem cells to replace the non-functional cells / form new populations of functional cells. (1)
By the eighth month, the number of white blood cells started to level off because the normal level had been reached. (1)

(Total: 5 marks)

13. (a) A high phosphate content in water promotes the growth of algae, causing excessive algal growth (algal blooms). (1)
Massive algal growth on water surfaces consumes large amounts of oxygen at night. (1)
This reduces dissolved oxygen levels in the water, and many aquatic organisms die from suffocation. (1)

(b) Egg cells and sperms are obtained from pigs and are fertilized *in vitro*. (1)
The gene coding for phytase is inserted into (the pronuclei of) the fertilized eggs. (1)

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The fertilized eggs containing the gene coding for phytase are cultured and developed into embryos. (1)

The embryos are implanted into the uterus of surrogate pigs. The surrogate pigs will give birth to offspring inserted with the gene coding for phytase. (1)

(c) Save costs in buying phytase supplement to mix with feed. /
Save expenses in treating pig waste before discharge. /
The GM pigs can obtain more nutrients from the same amount of feed and grow better.
(or other reasonable answers) (Any ***two***, 2)

(Total: 9 marks)

C. Exam challenge (8 marks)

14 (a) A monoclonal antibody is an antibody produced by a clone of cells developed from a single parent B cell. (1)

(b) Herceptin alone induces only slightly more cell death than control. The ability to induce cell death increases from 0.5 to 0.6 arbitrary units / has increased 0.1 arbitrary units. (1)

X-ray alone induces a little more cell death than Herceptin alone and control. The ability to induce cell death increases from 0.5 to 0.7 arbitrary units / has increased 0.2 arbitrary units. (1)

A combination of Herceptin and X-ray treatments induces much more cell death than either treatment alone; the ability to induce cell death increases from 0.5 to 2.0 arbitrary units / has increased 1.5 arbitrary units. (1)

(c) $(2.0 - 0.6) \div 0.6 \times 100\%$ (1)

$= 233\%$ increase (1)

(d) Herceptin identifies cancer cells by binding with the HER2 receptors on the surfaces of breast cancer cells. (1)

Herceptin helps immune cells to recognize breast cancer cells / attracts phagocytes and T cells to kill the cancer cells / other actions of antibody.

OR

It enhances the effect of X-ray in killing cancer cells. (1)

(Total: 8 marks)

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